HW1 Assessment Criteria

	Excellent	Good	Satisfactory	Unsatisfactory
Functionality **(20 pt.)**	The adder performs its specified function perfectly, passing more than 90% of the test cases. (**20-18 pt.**)	The adder mostly performs its specified function correctly, passing more than 75% of the test cases. **(17-15 pt.)**	The adder partially performs its specified function correctly, passing more than 50% of the test cases. **(14-10 pt.)**	The adder only partially fulfills its specified function, passing less than 50% of the test cases. **(9-0 pt.)**
Verilog Syntax **(5 pt.)**	Correctly implement adder algorithm and gate-level circuit structure. And the syntax is hardware-oriented encoding. **(5 pt.)**	Correctly implement adder algorithm and gate-level circuit structure. But only part of the syntax is hardware-oriented encoding. **(4-3 pt.)**	Correctly implement adder algorithm and gate-level circuit structure. But the syntax is software-oriented encoding. **(2-1 pt.)**	Only partially implement adder algorithm and gate-level circuit structure, the correctness is unsatisfied. **(0 pt.)**
Algorithm and Architecture **(15 pt.)**	Use multiple algorithms for comprehensive comparison and evaluation. Clearly explain the characteristics, advantages, disadvantage, operation methods, etc. of these algorithms. **(15-12 pt.)**	Use one kind of algorithm. Clearly explain its characteristics, advantages, disadvantages, operation methods, etc. **(11-8 pt.)**	Use one kind of algorithm. Only explain part of its characteristics, advantages, disadvantages, operation methods, etc, not complete enough. **(7-4 pt.)**	Use one kind algorithm. Briefly describe its characteristics, advantages, disadvantages, operation methods, etc, not complete enough and poor structure. **(3-0 pt.)**
Circuit Analysis **(15 pt.)**	Use multiple architectures for comprehensive comparisons and evaluations. Clearly analyze the number of logic gates, the number of logic gate layers, the critical path **(15-12 pt.)**	Use one kind of architectures. Clearly analyze the number of logic gates, the number of logic gate layers, the critical path indication, etc. of this algorithms. **(11-8 pt.)**	Use one kind of architectures. Only explain part of its number of logic gates, the number of logic gate layers, the critical path indication, etc., not complete enough. **(7-4 pt.)**	Use one kind of architectures. Briefly explain part of its number of logic gates, the number of logic gate layers, the critical path indication, etc., not complete enough **(3-0 pt.)**
Testbench and completeness of verification **(15 pt.)**	The verification and waveforms are clear and complete, considering various corner cases. The verification signal generation and comparison in testbench are performed automatically. **(15-14 pt.)**	The verification and waveforms are clear and complete, considering various corner cases. However, the verification signal generation and comparison in testbench are performed manually. **(13-11 pt.)**	Complete the basic verification, perform the waveform. However only consider part of corner cases. **(10-7 pt.)**	Only partial verification and wave pattern display are performed. The completeness is not good. **(6-0 pt.)**
Discussion **(20 pt.)**	Clearly explain the problems and solutions encountered during the design process, And reflect on the impact on the design **(15-14 pt.)**	Clearly explain the problems encountered during the design process and their solutions, but there is no reflection. **(13-11 pt.)**	It only describes some of the problems encountered during the design process and solutions. Not complete enough. **(10-7 pt.)**	Only briefly state the problems encountered during the design process and solutions. Poor structure. **(6-0 pt.)**
Report Overall **(10 pt.)**	Reports are clear and easy to read. Fonts, font styles, line spacing, indentations, pictures, etc. are neat and The paragraph hierarchy is clear and distinct. **(10-9 pt.)**	Reports are clear and easy to read.Fonts, font styles, line spacing, indentations, pictures, etc. are neat and However, the paragraph hierarchy is no clear enough. **(8-6 pt.)**	Only some fonts, font styles, line spacing, indentations, pictures, etc. of the report are neat. The paragraph hierarchy is not complete enough. **(5-3 pt.)**	The fonts, font styles, line spacing, indentations, pictures and other formats in the report are The paragraph hierarchical structure is **(2-0 pt.)**